

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 4: Heat

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	4 — Heat
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps baited here: equal-amount mixing must land midway while unequal amounts pull the mix toward the larger share; 'hotter = more heat' and 'feels colder = is colder' reflexes; division-size misreads where the small marks are NOT tenths; conductor/insulator role-swaps; emitter-vs-absorber colour confusion; breeze-direction reversals tied to which surface heats or cools faster. Do not write on this paper — mark answers on your answer sheet.

1. 200 g of water at 20 °C is mixed with 200 g of water at 80 °C in an insulated jar (no heat escapes). Using only the idea that equal amounts share the gap equally, the final temperature is closest to:

- (A) 80 °C, because the hotter water decides the final temperature
- (B) 100 °C, because the two temperatures add together (20 + 80)
- (C) 50 °C, midway between 20 and 80, because equal amounts of the same liquid meet halfway
- (D) 20 °C, because heat always settles at the lower of the two values

2. 400 g of water at 70 °C is mixed with only 100 g of water at 20 °C in an insulated jar. Without any formula, the soundest estimate of the final temperature is:

- (A) Exactly 45 °C, the midpoint of 20 and 70, regardless of the amounts
- (B) Below 45 °C, nearer 20 °C, because cold water always wins
- (C) Exactly 55 °C, because you average the two temperatures and then add ten
- (D) Above 45 °C — much nearer 70 °C — because the larger 400 g of hot water dominates the mix

3. A tiny drop of boiling water at 100 °C falls into a large bucket of water at 25 °C. The student must predict the bucket's final temperature. The best reasoning is:

- (A) It stays almost exactly 25 °C, because the bucket holds far more water and a single drop carries very little heat
- (B) It rises to about 62.5 °C, the midpoint of 25 and 100
- (C) It rises sharply because the drop is the hottest thing present
- (D) It drops below 25 °C because adding water always cools a container

4. Block X has twice as much metal as block Y, and both are the same metal. X is at 40 °C and Y is at 40 °C. They are pressed together. What happens to their temperatures?

- (A) X cools and Y warms, because the bigger block always gives heat to the smaller one
- (B) Nothing changes — both stay at 40 °C, because there is no temperature difference to drive any heat flow
- (C) Y warms toward X because more metal means a higher temperature
- (D) Heat flows from X to Y because larger objects hold more heat and must release it

5. A cup of tea is left in a room at 25 °C and slowly cools. A student says, 'The room is warming up the tea a little while the tea cools.' What is the correct description of the energy flow?

- (A) Heat flows both ways at once, so neither the tea nor the room changes
- (B) Cold flows from the room into the tea, which is what cools the tea
- (C) Heat flows from the hot tea into the cooler room air; the tea loses heat and the room gains a tiny amount, until the tea reaches room temperature
- (D) The room heats the tea because larger spaces always pass heat to smaller objects

6. Equal amounts of water are warmed: sample P from 20 °C to 30 °C, and sample Q from 70 °C to 80 °C. Each gained 10 °C. Which statement is the most accurate?

- (A) Q took in more heat, because it ended at the higher temperature
- (B) P took in more heat, because starting cold means it needs more heat to warm
- (C) Each sample took in roughly the same amount of heat, since equal amounts of the same liquid rose by the same number of degrees
- (D) No heat entered either sample, because heat and temperature are unrelated

7. A clinical thermometer reads only about 35 °C to 42 °C, while a laboratory thermometer reads about –10 °C to 110 °C. A technician must (i) record the freezing point of a salt solution near –5 °C and later (ii) note a feverish child's temperature to a tenth of a degree and carry the reading to another room. The correct pairing is:

- (A) Use the clinical thermometer for both tasks, since it reads in Celsius
- (B) Use the laboratory thermometer for both tasks, since its wide range covers everything

- (C) Use the laboratory thermometer for the -5°C salt solution and the clinical thermometer for the carried-away fever reading
- (D) Use the clinical thermometer for the -5°C solution and the laboratory thermometer for the fever

8. A clinical thermometer left reading 41.0°C from a very sick patient is wiped clean but not shaken down, then placed in a glass of warm water at 38°C . What does it show, and what should have been done?

- (A) About 41.0°C , because the kink holds the thread at the highest value reached and 38°C is not warm enough to push it higher — it should have been shaken down first
- (B) About 38°C , because the thermometer resets itself to each new use
- (C) About $41 + 38^{\circ}\text{C}$, because the two readings add together
- (D) About 38°C , because the thread always falls back to room temperature between uses

9. Why would a laboratory thermometer be a poor choice for confirming a fever of 39.2°C , even though 39.2°C lies well inside its -10°C to 110°C range?

- (A) A fever of 39.2°C is above its range, so it cannot reach that temperature
- (B) Its scale is spread over a huge range, so its smallest divisions are coarse (about 1°C) and it cannot resolve a tenth of a degree the way a clinical thermometer can
- (C) It has no kink, so it cannot measure body temperature at all
- (D) Its liquid freezes near body temperature, giving a wrong reading

10. A clinical thermometer and a laboratory thermometer are each warmed to 41°C and then laid on a cool table. A minute later, which still shows 41°C ?

- (A) Both still show 41°C , because every thermometer traps heat in its glass
- (B) Only the laboratory thermometer, because having no kink lets it hold its reading
- (C) Only the clinical thermometer — its kink keeps the thread at 41°C , while the kink-free laboratory thread slides down as its liquid cools on the table
- (D) Neither, because both reset to room temperature the instant they are removed

11. Thermometer P spans 0°C to 50°C with 100 small divisions over its glass; thermometer Q spans the same 0°C to 50°C over the same length but with only 25 small divisions. Which can be read more finely, and what is each smallest division worth?

- (A) Thermometer Q reads more finely, because fewer divisions each cover a smaller temperature
- (B) Both read equally finely, because they cover the same $0-50^{\circ}\text{C}$ range
- (C) Thermometer P's divisions are 0.5°C and Q's are 1°C
- (D) Thermometer P reads more finely; each of its divisions is 0.5°C , while each of Q's is 2°C

12. A clinical thermometer has 10 equal small divisions between each whole degree. A patient's thread stops on the 3rd small division above the 40 mark. The reading is:

(A) 40.03 °C

(B) 40.3 °C

(C) 43 °C

(D) 40.15 °C

13. On a laboratory thermometer the numbered marks are 50 degrees apart, with 10 equal small divisions between consecutive numbers. The thread rests on the 3rd small division above the 50 mark. The temperature is:

(A) 53 °C

(B) 65 °C

(C) 50.3 °C

(D) 100 °C

14. A laboratory thermometer is numbered every 10 degrees with 5 equal divisions between numbers. In the morning the thread sat on the 2nd division above the 20 mark; by noon it sits on the 4th division above the 30 mark. By how much did the temperature change?

(A) It rose by 4 °C, from 22 °C to 26 °C

(B) It rose by 14 °C, from 24 °C to 38 °C

(C) It rose by 10 °C, the gap between the 20 and 30 marks

(D) It rose by 2 °C, because the division count rose from 2 to 4

15. A laboratory thermometer is numbered every 5 degrees with 5 equal divisions between numbers. The thread sits on the 3rd division below the 0 mark (in the negative region). What temperature does it read?

(A) 3 °C

(B) -15 °C

(C) -8 °C

(D) -3 °C

16. On a laboratory thermometer the numbered marks are 3 degrees apart with 6 equal divisions between them. The thread is on the 5th division above the 12 mark. One student writes 12.5 °C, the other writes 14.5 °C. Who is right and why?

(A) 12.5 °C is right, because 5 divisions above a mark always means five-tenths

(B) Both are right, since 12.5 and 14.5 are just two ways of writing the same value

(C) 14.5 °C is right — each division is 0.5 °C, so 5 divisions above 12 is $12 + 2.5 = 14.5$ °C

(D) Neither is right; 5 divisions above 12 is 17 °C

17. A laboratory thermometer's thread climbs steadily during heating. At 9:00 it read the 2nd division above 30; at 9:01 the 5th division above 30; at 9:02 the 8th division above 30. Each division is 1 °C. If the same steady rate continues, what will it read at 9:03?

(A) About 39 °C, because the thread cannot rise past the 9th division above 30

(B) About 48 °C, because the temperature doubles each minute

(C) About 41 °C, since it rises 3 °C each minute (32, 35, 38, then 41 at 9:03 — i.e. the 1st division above 40)

(D) Exactly 40 °C, because the thread always stops at a numbered mark

18. A metal spoon's bowl rests in a pot of hot soup, while tiny dabs of butter are stuck along the handle at equal spacing. The dabs melt one after another — the dab nearest the bowl first and the farthest one last, each a little later than the one before. The clearest conclusion about how heat moves AND why the timing is gradual is:

- (A) Heat radiates equally to every dab at once, so they should all melt together — the gradual order is just chance
- (B) Hot metal physically rises and carries heat dab to dab, which is convection inside the solid handle
- (C) Each dab makes its own heat once it warms, so the order shows the dabs igniting in turn
- (D) Heat conducts particle-to-particle along the handle, arriving later at points farther from the hot bowl, so distant butter melts later

19. Four identical rods — aluminium, iron, glass, and plastic — are coated in candle wax and heated equally at one end. The wax clears from the aluminium rod first, then the iron, then the glass, while the plastic rod's wax hardly melts at all. Ranking the four from best to worst heat conductor, the correct order is:

- (A) Plastic, then glass, then iron, then aluminium, because lighter materials carry heat fastest
- (B) Iron, then aluminium, then plastic, then glass, because iron is the best conductor of the four
- (C) Aluminium, then iron, then glass, then plastic — the rod whose wax clears soonest conducts heat fastest
- (D) All four conduct equally well, since the heat source was identical for each

20. A steel skewer and a bamboo skewer of identical shape are both heated at one end. To make the far end of the bamboo skewer reach the same temperature the steel skewer's far end reached in 1 minute, the bamboo skewer must be heated:

- (A) For exactly 1 minute, since both skewers are the same shape and size
- (B) For less than 1 minute, because poor conductors heat up faster
- (C) For much longer than 1 minute, because bamboo is a poor conductor and passes heat to its far end far more slowly
- (D) Never, because bamboo cannot conduct any heat at all

21. A ceramic floor tile and a steel knife have both been resting on the same kitchen counter and are at the same 18 °C. You press your warm palm onto each. From which does your hand lose heat faster, and which feels colder?

- (A) The ceramic tile feels colder, because ceramic is at a lower temperature than the steel
- (B) Your palm loses heat faster to the steel knife, so the steel feels colder, even though both are at 18 °C
- (C) Both feel equally cold, since both are at 18 °C
- (D) The steel feels warmer, because metal stores more heat and gives some back to the hand

22. Heat passes quickly along a solid iron bar but barely creeps through the still air trapped inside a woollen sweater. In terms of particles, why?

- (A) The wool's trapped air has no particles, so there is nothing in it to carry heat by conduction
- (B) Iron is colder than air, and colder materials always conduct better
- (C) The air particles move too slowly to carry any energy, so conduction stops
- (D) Iron's particles are packed closely and pass on vibrations to neighbours readily, while the trapped air's particles are far apart and rarely collide, so they hand on energy very slowly

23. An electric kettle must boil its water quickly from the heating element, yet have a grip that stays cool to hold. Which combination of materials achieves BOTH goals?

- (A) A metal body (good conductor, passes heat into the water) with a plastic grip (poor conductor, keeps heat from reaching the hand)
- (B) A plastic body and a metal grip, so heat is kept out of the water and led safely along the grip
- (C) A metal body and a metal grip, since one material is simplest and metal grips stay coolest
- (D) A plastic body and a plastic grip, so the whole kettle resists heat and never gets dangerously hot

24. A pot of soup is heated only at its base. The layer of soup right at the bottom, once warmed:

- (A) Becomes denser and sinks, while cool soup rises over it, because warmed liquid is always the heaviest
- (B) Stays still and simply conducts its heat straight up through the layers above without moving
- (C) Turns instantly to vapour at the bottom and shoots straight up without any circulation
- (D) Expands, becomes less dense, and rises, while cooler denser soup sinks to take its place, setting up a circulating current

25. A wide flat pan of water is heated by a small flame placed under its left edge only. Sawdust sprinkled in the water shows where the convection currents form. What does it reveal?

- (A) The current always rises at the pan's centre, no matter where the flame is
- (B) There is no current; the sawdust only spreads by conduction through the still water
- (C) The sawdust sinks first over the flame because heating makes water heavier there
- (D) The current rises directly over the heated left edge and sinks on the cooler far side, so an off-centre flame still drives a loop rooted at that edge

26. A family mounts their room heater high on a wall near the ceiling and complains the floor stays cold; moving it down near the floor fixes the problem. Why?

- (A) A ceiling-level heater heats the room faster, because heat from up high falls down to fill the room
- (B) Position makes no difference, because air carries heat equally well from any height by conduction
- (C) With a low heater the floor becomes the hottest part, because warm air always sinks to the floor
- (D) Warm air from a ceiling-level heater stays near the ceiling, leaving the lower room poorly heated; a low heater drives a convection loop that warms the whole room — which is why heaters go low

27. A student insists, 'If I heat a brick wall in its middle, the warm part will rise and the cold parts will sink, just like water in a kettle, setting up a convection current inside the brick.' Why is this impossible?

- (A) It is possible, but only if the brick is first painted black to absorb the heat
- (B) It is possible, because any material can convect once it is hot enough
- (C) Convection needs a fluid whose heated portions can physically move and circulate; in a solid brick the particles are fixed in place and cannot flow, so heat moves only by conduction
- (D) It is impossible because a brick cannot be heated in the middle at all

28. A beaker of water is heated by a coil placed near the TOP of the water rather than at the bottom. The water lower down stays cold for a long time, and almost no convection current develops. The best explanation is:

- (A) Heating at the top leaves the warm, lighter water on top of the denser cold water below — already the stable arrangement — so there is no buoyancy difference to drive circulation, and heat only creeps down by slow conduction
- (B) Warm water is denser, so heating the top should send it to the bottom and stir the beaker
- (C) Convection cannot occur in any beaker, only in wide flat pans
- (D) The cold water at the bottom blocks the warm water above from ever cooling

29. On a calm sunny day, a hot-air-balloon pilot looks for 'thermals' — columns of rising warm air. Over which surface is the strongest thermal most likely to form, and why?

- (A) Over a shaded pond, because water is always the warmest surface
- (B) Over a dark ploughed field, which absorbs more sunlight and warms the air above it more, so that air rises faster as a stronger upward convection current
- (C) Over a white concrete lot, because pale surfaces push air upward
- (D) Equally over every surface, since the same sunlight falls on all of them

30. You stand in front of a glowing electric room heater. The hot air above its coils streams straight up to the ceiling, yet your face feels warm, and a clear glass screen placed in front still lets warmth through. The warmth reaches you mainly by:

- (A) Radiation, which travels in straight lines, needs no moving air, and can pass through transparent glass to reach your skin
- (B) Convection, because the hot air must be curling around the screen to reach your face
- (C) Conduction, through a chain of touching air running from the coil to your face
- (D) The heater pushing the cold room air toward you through the glass

31. The same iron poker is held in a fire until it glows: first a dull red (about 600 °C), and later a bright orange (about 900 °C). At which stage does it give off more heat by radiation each second, and why?

- (A) At 600 °C (dull red), because cooler objects shed their heat faster to catch up
- (B) At 900 °C (bright orange), because the hotter an object is, the more radiation it emits each second
- (C) The same at both stages, because it is the one same poker
- (D) Neither, because iron only radiates in sunlight

32. Inside a laboratory vacuum chamber, a hot metal sample hangs from a thin insulating thread; even after all the air is pumped out, the sample still slowly cools. By which mode does it lose this heat, and why are the others ruled out?

- (A) By convection, since hot gas streams off the sample into the chamber
- (B) By conduction, since the vacuum directly carries heat away from the surface
- (C) It loses heat by radiation, which needs no medium; with the air pumped out there are almost no particles to conduct through and no fluid to convect, so only radiation remains
- (D) It cannot lose any heat at all, because all three modes need matter

33. Two identical mugs of equally hot coffee sit in the same cool room: mug P has a matt-black surface; mug Q is wrapped in shiny foil. After ten minutes, which coffee is cooler, and why?

- (A) Mug P (matt black) is cooler, because dull dark surfaces are better emitters of radiation and lose heat faster
- (B) Mug Q (shiny foil) is cooler, because shiny surfaces radiate heat away fastest
- (C) Both are equally cool, because they hold the same coffee at the same start temperature
- (D) Neither cools by radiation indoors, because radiation needs direct sunlight

34. A matt-black water tank on a roof both heats up faster in the morning Sun AND cools down faster after dark than an identical shiny tank of the same size. Which single principle explains BOTH observations?

- (A) Dull dark surfaces are both better absorbers AND better emitters of radiation, while shiny surfaces are poorer at both — so the black tank gains heat faster in the Sun but also loses it faster after dark
- (B) Shiny surfaces both absorb and emit radiation better, which is why the shiny tank changes faster
- (C) Dark surfaces absorb well but cannot emit, while shiny surfaces emit well but cannot absorb
- (D) Colour affects only absorption, never emission, so the cooling difference must be due to something else

35. Across the empty gap between the double walls of a vacuum flask, which of the three modes of heat transfer can still carry heat, and how many?

- (A) Exactly two, because conduction and radiation both cross the gap but convection cannot
- (B) All three, because heat always finds some way across a gap
- (C) Exactly one — only radiation, since it needs no medium, while conduction needs touching particles and convection needs a flowing fluid, both absent in the vacuum
- (D) None, because every mode of heat transfer needs matter to travel through

36. A soldering iron needs a fixed handle a worker can hold while its tip reaches hundreds of degrees. From the materials brass, copper, steel, and cork, which is correct for the handle, and why are the others wrong?

- (A) Cork, because it is a poor conductor and stays cool to grip; brass, copper, and steel are all metals that conduct heat well and would get hot
- (B) Brass, because as a good conductor it spreads the heat out and so stays coolest
- (C) Steel, because metals barely conduct heat and stay cool
- (D) Copper, because metal handles are the most comfortable and durable

37. A double-glazed window traps a thin near-still-air gap between two glass panes and carries a shiny low-emissivity (mirror-like) coating on one pane. Match each feature to the mode of heat loss it chiefly fights:

- (A) The trapped-air gap fights conduction and convection (both need matter to cross it), while the shiny low-emissivity coating cuts radiation (shiny surfaces are poor emitters)
- (B) The gap fights radiation, while the shiny coating fights conduction and convection
- (C) The gap blocks all three modes by itself, and the shiny coating only improves the appearance
- (D) The shiny coating blocks all three modes, and the gap only adds strength

38. A glass tumbler, a foam box, and an aluminium can of the same shape are each filled with the same hot soup in the same room. Ranking how well they keep the soup hot, from best to worst, the most reasonable order is:

- (A) Aluminium can best, foam worst, because metal seals heat in by surrounding the soup
- (B) All three keep it equally hot, since they hold the same soup in the same room
- (C) Foam box best, glass tumbler in the middle, aluminium can worst — foam's trapped air is the poorest conductor and the metal can conducts heat away fastest
- (D) Glass best, foam worst, because solid walls block heat better than air pockets

39. A cyclist feels comfortably warm in a knitted jumper while standing still, but suddenly cold when riding fast into the wind, even though the jumper is unchanged. Why?

- (A) The jumper's air is a perfect insulator whether still or moving, because it has so few particles
- (B) Still trapped air in the jumper conducts heat poorly and insulates well, but the wind of riding moves the air and carries heat away by convection, so it no longer insulates
- (C) Moving air conducts heat better than still air, which is why the wind cools the rider
- (D) The jumper destroys heat when the air is still but creates heat when it moves

40. A marathon runner is given a thin shiny 'space blanket' at the finish to avoid chilling. Unlike a thick wool blanket which works by trapping air, the shiny sheet keeps the runner warm chiefly by:

- (A) Generating warmth from the shiny coating once it touches the skin
- (B) Trapping a very thick layer of air inside the thin shiny sheet
- (C) Conducting the cold surroundings away from the runner quickly
- (D) Reflecting the runner's own radiated body heat back toward them instead of letting it escape

41. Two people on a sunny beach wear shirts identical except for colour — one white, one black. The same two shirts are then worn at night facing a warm bonfire. In daytime sun the black-shirt wearer feels hotter; at night near the fire the black-shirt wearer ALSO loses body heat faster if they turn away from the fire. Which single fact explains both?

- (A) Black absorbs well in the day but reflects well at night, switching its behaviour with the time
- (B) White absorbs more during the day and emits more at night, so the colours swap roles
- (C) Dark surfaces both absorb radiation better (so they soak up the Sun) and emit radiation better (so they shed body heat faster), while white reflects and emits poorly
- (D) Colour only affects daytime comfort; the night-time difference must come from the cloth's thickness

42. A camper paints the outside of a water bag dull black and hangs it in the Sun to warm water for an evening wash. Why dull black rather than shiny silver?

- (A) A dull black surface reflects the most sunlight, sending heat into the water by reflection
- (B) A shiny silver surface would absorb more, but black is chosen because it is cheaper

- (C) Colour does not matter for the bag; the Sun heats the water regardless
- (D) A dull black surface absorbs the most of the Sun's radiation, heating the bag and the water inside most effectively

43. Two identical garden water-storage drums stand in full Sun all afternoon — one painted white, one painted dark brown, both the same size and thickness. By evening, which drum's water is cooler, and why?

- (A) The white drum's water is cooler, because its light surface reflects most of the Sun's radiation, so the walls absorb less heat
- (B) The dark brown drum's water is cooler, because dark colours keep things cool by absorbing heat away
- (C) Both are equally cool, because the drums are the same size and thickness
- (D) The white drum is hotter, because white absorbs the most sunlight

44. A trekker at high altitude in bright midday Sun must choose a hat — black or white, otherwise identical. Which keeps the head warmer in the Sun, and what is the trade-off at night?

- (A) The white hat is warmer in the Sun, because pale colours absorb more sunlight
- (B) The black hat is warmer in the midday Sun because it absorbs more solar radiation; but at night, away from the Sun, the black hat would also emit the head's heat away faster
- (C) The black hat is warmer at every time, because black always makes more heat
- (D) Colour makes no difference in the Sun, since both hats are the same thickness

45. Two solar cookers, one lined dull black inside and one lined shiny silver, sit in identical bright sunlight under glass lids; after an hour the black-lined one is far hotter. If both are then carried into a shaded cool shed, which cooker's thermometer falls fastest, and why?

- (A) The shiny silver cooker cools fastest, because shiny surfaces lose heat by radiation fastest
- (B) Both cool at the same rate, because they are the same size and shape
- (C) Neither cools, because once heated a closed cooker keeps its heat
- (D) The black-lined cooker cools fastest, because a dull dark surface is also a better emitter and radiates its heat away faster than the shiny one

46. At 3 p.m. on a sunny coast, beach flags stream steadily from the water toward the shops behind the beach. Combining the daytime breeze pattern with its cause, this is a:

- (A) Land breeze: the sea heats faster than the land by day, pushing air toward the shore
- (B) Sea breeze caused by the Sun directly dragging air landward across the water
- (C) Land breeze caused by the land staying cooler than the sea all afternoon
- (D) Sea breeze: the land heats faster than the sea by day, so air rises over the warm land and cooler air flows in from the sea

47. At about 2 a.m., fishermen launch their boats and are carried gently out to sea by a steady offshore wind. Identify the breeze and the temperature situation that drives it:

- (A) A sea breeze: the sea has cooled faster than the land, so air flows toward the open sea
- (B) A land breeze: the land stays warmer than the sea at night, pushing air seaward
- (C) A sea breeze caused by the night air automatically reversing the daytime wind
- (D) A land breeze: the land has cooled faster than the sea, so the sea is now warmer; air rises over the warm sea and cooler air flows out from the land

48. A kite-surfer finds the coastal wind reliably strong by mid-morning and again after sunset, but it dies away to almost nothing for a while around sunrise and once more near sunset. The calm, shifting periods happen because:

- (A) Near sunrise and sunset the land and sea are at nearly the same temperature, so there is little difference to drive air in either direction
- (B) The wind always stops for exactly one hour twice a day by a fixed rule
- (C) The Sun is directly overhead at sunrise and sunset, cancelling all breezes
- (D) The land–sea temperature difference is at its greatest at sunrise and sunset

49. Two students debate the coastal winds: one says the day and night breezes are unrelated phenomena driven by different forces; the other says they are 'one convection process working both ways.' Why is the second student correct?

- (A) Both are convection: air rises over whichever surface is warmer and cooler air flows in from the cooler surface; by day the land is warmer, by night the sea is warmer, so the same process simply reverses
- (B) The day breeze is convection but the night breeze is conduction through the cooler ground
- (C) The day breeze is driven by the Sun and the night breeze by the Moon, so they are different forces
- (D) They differ because the day breeze is radiation and the night breeze is convection

50. Imagine a foggy, overcast coastal day on which the land and sea happen to hold exactly the same temperature all day long. Would a sea breeze or a land breeze blow? Why?

- (A) A sea breeze, because the sea is always cooler than the land regardless of temperature
- (B) Neither — with no temperature difference, no surface is warmer to make air rise, so there is nothing to drive a breeze in either direction
- (C) Both at once, because the coast always produces some breeze
- (D) A land breeze, because air always flows out to sea on overcast days

51. Sea and land breezes are convection on a large coastal scale. In this convection, the rising air over the warmer surface is replaced by air flowing in along the surface from the cooler region. This means that, at any moment, the breeze you feel near the ground always blows:

- (A) From the warmer surface toward the cooler one, following the rising warm air
- (B) From the cooler surface toward the warmer one (sea to land by day, land to sea by night), filling in beneath the rising warm air
- (C) Always from the sea toward the land, both day and night
- (D) Straight upward only, since warm air rises vertically

52. A quilted jacket — cloth stitched into many puffy pockets — keeps a child warmer than a flat jacket made of the same weight of the same cloth. The main reason the quilting is warmer is:

- (A) The quilted jacket weighs more, and heavier clothing is always warmer
- (B) Each puffy quilted pocket traps an extra pocket of still air, and trapped still air is a poor conductor, so more trapped-air pockets slow heat loss more
- (C) The quilting makes the cloth produce more body heat
- (D) The quilted cloth conducts the body's heat outward faster, which feels warm

53. A down sleeping bag squashed flat under a heavy weight keeps a camper far less warm than the same bag fluffed up, even though it contains exactly the same amount of down. Why?

- (A) Fluffing the down traps thick pockets of still air, the real insulator; squashing it flat removes the trapped air, so heat escapes faster
- (B) Squashing the down stops it producing heat, so the bag goes cold
- (C) The squashed down conducts the body's heat away faster because it is now thicker
- (D) A flat bag reflects less body heat, so the camper loses heat by reflection

54. To keep a pot of cooked rice warm on the table, a person can either drape it in a thick towel or stand it on a sheet of shiny metal. Which keeps the rice warm longer, and why?

- (A) Drape it in the thick towel, because the towel is full of trapped still air and is a poor conductor, so it greatly slows the rice's heat from escaping
- (B) Stand it on the shiny metal sheet, because metal seals heat in by blocking it
- (C) Use only the metal sheet, because thin metal insulates better than a thick towel
- (D) Either choice works equally well, since both feel warm when held

55. Order these four mugs of identical hot tea, in the same room, from the one that stays hot LONGEST to the one that cools soonest: (i) a bare ceramic mug, (ii) a foam mug with a tight lid, (iii) a bare thin metal mug, (iv) a ceramic mug wrapped in a thick towel. The correct order is:

- (A) Bare thin metal mug longest, then bare ceramic mug, then towel-wrapped ceramic, then foam-with-lid soonest
- (B) Foam mug with a lid longest, then towel-wrapped ceramic mug, then bare ceramic mug, then bare thin metal mug soonest

- (C) All four cool at the same rate, since they hold the same tea in the same room
(D) Bare ceramic mug longest, then foam-with-lid, then bare metal, then towel-wrapped ceramic soonest

56. A student labels three everyday cases and the mode involved: 'A metal ladle left standing in a boiling pot becomes hot to hold — conduction. Standing in sunlight coming through a closed window feels warm on the skin — radiation. Water bubbling and churning as it boils on a flame — convection.' How many of these three pairings are correct?

- (A) All three are correct
(B) Only two are correct
(C) Only one is correct
(D) None is correct

57. A picnic cold-box keeps ice for most of a day, but the ice always melts eventually, even though the box has thick foam walls and a sealed lid. Which statement best explains why a cold-box only SLOWS the warming rather than stopping it forever?

- (A) A cold-box makes a little cold of its own, which runs out after a day
(B) The foam actually speeds up heat entry, so the ice melts as fast as in the open air
(C) Cold from the room flows into the box through the foam, which a cold-box cannot block
(D) The thick foam and sealed lid greatly slow conduction, convection, and radiation, but heat still leaks slowly through the walls, the lid seams, and every opening, so the box cannot be perfect

58. Consider a charcoal grill warming a cook standing beside it on a calm evening: (1) the glowing coals warm the cook's face across the gap without touching it; (2) the air directly above the coals is heated and rises; (3) cooler room air slides in along the floor toward the grill. Identify the heat-transfer mode in step 1 and in steps 2–3, in that order:

- (A) Step 1 is conduction (the coals conduct heat to the face); steps 2–3 are radiation (the warm air beams to the cook)
(B) Step 1 is convection (hot gas reaches the face); steps 2–3 are conduction (the air conducts heat sideways)
(C) Step 1 is radiation (heat crossing the gap to the face); steps 2–3 are convection (warm air rising and cooler air flowing in)
(D) All three steps are conduction, since the coals, air, and cook are all in contact

59. A laboratory thermometer reads 28 °C on a desk. A student cups both hands tightly around the bulb until the thread rises to about 34 °C, then opens their hands and watches. What does the thread do, and what does this show about how the thermometer works?

- (A) The thread stays at 34 °C, because a thermometer always keeps its highest reading
(B) The thread keeps rising past 34 °C, because the hands keep adding heat after opening

- (C) The thread drops to 0°C , because a thermometer with no hand on it reads zero
- (D) The thread falls back toward 28°C , showing the liquid expands when warmed by the hands and contracts again as it cools, with no kink to hold it

60. On a cold morning a marble step and a wooden step outside a house have both reached the air temperature of 5°C . A barefoot person steps briefly on each. Which feels colder underfoot, and would a thermometer pressed on each read differently?

- (A) The wood feels colder, and a thermometer would read a lower temperature on the wood
- (B) The marble feels much colder underfoot because stone conducts heat from the foot quickly, yet a thermometer would read 5°C on both since they are at the same temperature
- (C) Both feel equally cold, and a thermometer reads 5°C on both, since feel and temperature always match
- (D) The marble feels colder, and a thermometer would read a lower temperature on the marble, confirming it is genuinely colder

Solutions — Science (Class 7), Chapter 4: Heat — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	B	31	B	41	C	51	B
2	D	12	B	22	D	32	C	42	D	52	B
3	A	13	B	23	A	33	A	43	A	53	A
4	B	14	B	24	D	34	A	44	B	54	A
5	C	15	D	25	D	35	C	45	D	55	B
6	C	16	C	26	D	36	A	46	D	56	A
7	C	17	C	27	C	37	A	47	D	57	D
8	A	18	D	28	A	38	C	48	A	58	C
9	B	19	C	29	B	39	B	49	A	59	D
10	C	20	C	30	A	40	D	50	B	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Heat flows from the hotter water to the cooler water until both reach one common temperature. With equal amounts of the same liquid, that common value sits exactly halfway between the two starting temperatures: $(20 + 80) \div 2 = 50$ °C. The hotter portion does not simply impose its 80 °C — it cools as it warms the other half. Adding the two temperatures (giving 100) treats temperature like a quantity that piles up, which it does not. And heat does not settle at the lower value of 20 °C; the cool water warms up while the hot water cools down.

2. (D) When the amounts are unequal, the final temperature is pulled toward the temperature of the larger amount. Here there is four times as much hot water (400 g) as cold water (100 g), so the mix ends up well above the midpoint of 45 °C and much closer to 70 °C. The midpoint of 45 °C would only be right if the amounts were equal. 'Cold water always wins' is false — the side with more substance has the bigger say, and here that is the hot water. Plainly averaging the two temperatures and nudging the result also ignores that the larger hot share controls the outcome.

3. (A) The final temperature is dragged toward whichever side has far more substance. The bucket has overwhelmingly more water than one drop, so the bucket barely changes and stays near 25 °C; the drop's small heat is shared out among a huge mass of water. The midpoint of 62.5 °C would only apply if the two amounts were equal, which they are wildly not. Being the hottest object does not let one drop heat a whole bucket — heat content depends on amount as well as temperature. And adding warm water cannot cool the bucket below 25 °C.

4. (B) Heat flows only when there is a temperature difference, always from hotter to colder. Both blocks are at 40 °C, so there is no difference and no net flow — both simply stay at 40 °C. Having more metal gives block X more total heat, but the same temperature, so it does not push heat into Y. 'More metal means higher temperature' confuses amount of heat with temperature; they are different. And although X stores more heat than Y, heat does not flow without a temperature gap.

5. (C) Heat travels from the hotter body to the cooler one. The tea is hotter than the room, so heat moves out of the tea into the room air; the tea cools and the room warms by a tiny, barely noticeable amount, until the tea matches room temperature. The student's claim reverses this — the room does not warm the tea. There is no separate substance called 'cold' that flows; cooling is just the loss of heat. And size does not set the flow direction — temperature difference does.

6. (C) For the same liquid and the same amount, the heat needed depends on how many degrees the temperature rises, not on the starting value. Both rose by 10 °C, so each absorbed about the same heat. Ending at a higher temperature (Q at 80 °C) does not by itself mean more heat was added during this step. Starting cold does not make P need extra heat for the same 10 °C climb. And heat is exactly what raised each temperature, so saying no heat entered is wrong — heat and temperature change are tightly linked here.

7. (C) Task (i) is at -5 °C, far below the clinical scale's lowest mark of 35 °C, so only the wide-range laboratory thermometer can reach it. Task (ii) needs a reading held in place after the thermometer leaves the body and carried to another room — that is exactly what the clinical thermometer's kink does, plus its scale is spread finely over the body range for tenth-of-a-degree fevers. Using the clinical one for both fails on (i) because -5 °C is off its scale. Using the laboratory one for both fails on (ii) because its kink-free thread would slip as it cooled on the walk to the other room. The last pairing assigns the wrong tool to each task.

8. (A) The kink traps the thread at the highest temperature it has reached. After the sick patient it sits at 41.0 °C. The warm water at 38 °C is cooler than 41.0 °C, so the liquid does not expand enough to push the trapped thread any higher — it stays near 41.0 °C, giving a falsely high reading. This is precisely why a clinical thermometer must be shaken down before each use. The thermometer does not reset itself; that is the job of the shake.

Readings do not add up across uses. And the kink stops the thread from dropping back, so it does not fall to room temperature on its own.

9. (B) A laboratory thermometer must cover about 120 degrees of range, so each small division is comparatively large (around 1 °C), and it cannot distinguish 39.2 °C from 39.3 °C. A clinical thermometer devotes its whole length to roughly 35–42 °C, so it shows tenths of a degree clearly. The range claim is false: 39.2 °C is inside –10 to 110. Lacking a kink only means the reading is not held while carried away — it can still measure body temperature in contact. And the liquid does not freeze anywhere near body temperature.

10. (C) A clinical thermometer has a kink that traps the thread at the highest temperature reached, so once laid on the cool table it still reads 41 °C. A laboratory thermometer has no kink, so its thread moves freely with temperature; as its liquid cools on the table the thread slides back down, so it does NOT hold 41 °C. It is not true that glass traps heat to lock a reading — only the kink locks it. The kink-free laboratory thermometer is precisely the one that loses the reading, not the one that keeps it. And neither thermometer resets the instant it is removed; the clinical one in particular keeps its trapped reading.

11. (D) Across 50 °C, P's 100 divisions make each one $50 \div 100 = 0.5$ °C; Q's 25 divisions make each one $50 \div 25 = 2$ °C. The thermometer whose smallest division is smaller can be read more finely, so P (0.5 °C steps) beats Q (2 °C steps). Fewer divisions over the same range make each division LARGER, not smaller, so Q is coarser. Equal ranges do not guarantee equal fineness — division count decides it. And the computed values are 0.5 °C and 2 °C, not 0.5 °C and 1 °C.

12. (B) Ten equal divisions in one degree make each division $1 \div 10 = 0.1$ °C. Three divisions above 40 means $40 + 3 \times 0.1 = 40.3$ °C. Reading 40.03 places the decimal one digit too far, as if each division were 0.01 °C. Counting the 3 divisions as whole degrees ($40 + 3 = 43$) ignores that they are tenths. And 40.15 would require divisions of 0.05 each, which is not the stated scale.

13. (B) Fifty degrees split into 10 equal divisions makes each division $50 \div 10 = 5$ °C. Three divisions above 50 means $50 + 3 \times 5 = 50 + 15 = 65$ °C. Reading 53 wrongly treats each division as 1 °C ($50 + 3$). Reading 50.3 wrongly treats each division as 0.1 °C. And 100 jumps to the next numbered mark, ignoring that the thread is only 3 of the 10 divisions up.

14. (B) Ten degrees over 5 divisions makes each division 2 °C. Morning: $20 + 2 \times 2 = 24$ °C. Noon: $30 + 4 \times 2 = 38$ °C. The change is $38 - 24 = +14$ °C, a rise. Reading the divisions as 1 °C each (giving 22 and 26) halves the true values. The 10 °C gap between the numbered marks is not the change in the thread's position. And simply comparing the division counts (2 vs 4) and calling it a 2 °C rise ignores both the 2 °C size of each division and the move from the 20 mark up to the 30 mark.

- 15. (D)** Five degrees over 5 divisions makes each division $1\text{ }^{\circ}\text{C}$. Three divisions below 0, moving toward colder, gives $0 - 3 = -3\text{ }^{\circ}\text{C}$. Reading $+3\text{ }^{\circ}\text{C}$ ignores that the thread is below zero, in the colder direction. Reading -15 wrongly treats each division as $5\text{ }^{\circ}\text{C}$. And -8 wrongly starts counting from the -5 mark instead of from 0.
- 16. (C)** Three degrees split into 6 divisions makes each division $3 \div 6 = 0.5\text{ }^{\circ}\text{C}$. Five divisions above 12 means $12 + 5 \times 0.5 = 12 + 2.5 = 14.5\text{ }^{\circ}\text{C}$, so the second student is correct. The first student assumed each division was $0.1\text{ }^{\circ}\text{C}$, which is only true when there are 10 divisions per degree — not the case here. The two values 12.5 and 14.5 are clearly different numbers. And $17\text{ }^{\circ}\text{C}$ would require each division to be $1\text{ }^{\circ}\text{C}$ with five of them, which contradicts the stated $0.5\text{ }^{\circ}\text{C}$ divisions.
- 17. (C)** Each division is $1\text{ }^{\circ}\text{C}$, so the readings are $30 + 2 = 32$, $30 + 5 = 35$, $30 + 8 = 38\text{ }^{\circ}\text{C}$ — a steady rise of $3\text{ }^{\circ}\text{C}$ per minute. Continuing the pattern, 9:03 gives $38 + 3 = 41\text{ }^{\circ}\text{C}$, which is 1 division above the 40 mark. The thread is not capped at the 9th division above 30; it simply crosses into the next numbered region. The rise is steady (adding 3 each minute), not doubling. And there is no rule forcing the thread to stop on a numbered mark — it lands wherever the temperature is.
- 18. (D)** In a solid, heat travels by conduction: vibrating particles pass energy to their neighbours, so the disturbance reaches near points first and far points later. That is exactly why the butter melts in sequence from the bowl end outward along the handle — a gradual arrival, not chance. If radiation reached every dab equally and instantly, they would melt together, which they do not. Convection needs a fluid that can flow; the solid handle's particles are locked in place and cannot circulate. And the butter does not make its own heat — it merely melts when conducted heat arrives.
- 19. (C)** The faster the wax clears, the faster heat reached that point, so the best conductor loses its wax first. Aluminium clears first, then iron, then glass, while plastic barely melts — so the order is aluminium > iron > glass > plastic. Being light does not improve conduction; plastic is light yet the poorest conductor here. Iron conducts well but less than aluminium in this set, and the order must follow the melt times, not a guess. And an identical heat source does NOT equalise the results — the material decides how quickly heat passes through, which is the whole point of the experiment.
- 20. (C)** Same shape means the only difference is the material. Steel is a good conductor and quickly delivers heat to its far end; bamboo is a poor conductor, so heat creeps along it slowly and the far end needs much more time to reach the same temperature. Identical shape does not make the times equal, because the materials conduct at very different rates. Poor conductors heat their far ends MORE slowly, not faster. And bamboo does conduct heat — just slowly — so it is not 'never.'
- 21. (B)** Both objects are at $18\text{ }^{\circ}\text{C}$, so neither is actually colder. But the steel knife is a good conductor: it whisks heat away from your warmer palm quickly, cooling the skin fast, so it

FEELS colder. Ceramic is a poorer conductor and draws heat more slowly, so it feels less cold. The ceramic is not at a lower temperature — both are 18 °C. They do not feel equally cold despite equal temperature, because conduction rates differ. And the steel, being colder than your hand, takes heat from it rather than giving heat back, so it cannot feel warmer.

22. (D) Conduction is the passing of energy from particle to neighbouring particle. In iron the particles are tightly packed and constantly in contact, so energy is relayed quickly. In the air trapped in wool the particles are widely separated and seldom touch, so the relay is feeble and slow. The trapped air certainly contains particles — they are just spread out. Temperature does not set conduction ability; iron is not 'colder.' And air particles actually move fast; it is their large spacing, not slowness, that makes conduction poor.

23. (A) The body must carry heat from the heating element into the water, which calls for a good conductor — metal. The grip must keep heat from reaching the hand, which calls for a poor conductor (insulator) — plastic. Reversing them (plastic body, metal grip) would leave the water barely warmed while the grip conducts heat and burns the hand. An all-metal kettle would make the grip hot, not cool. And an all-plastic kettle would not heat the water well, since plastic is a poor conductor — defeating the boiling goal.

24. (D) Heated soup expands, so the same mass takes more space, making it less dense (lighter) — it rises. Cooler, denser soup sinks to replace it, and the cycle repeats, forming a convection current that warms the whole pot. Warmed liquid is lighter, not heavier, so it does not sink. Liquids conduct heat poorly, so the heat is not simply conducted straight up through still layers — the bulk movement of soup does most of the carrying. And the soup does not flash to vapour at ordinary heating; it circulates as a liquid first.

25. (D) Convection currents rise wherever the water is heated, because that is where the water expands, lightens, and floats upward; the cooler water elsewhere sinks. Heating the left edge therefore anchors the rising limb at that edge, not the centre. The current does not always rise centrally — it follows the heat source. Conduction through still water is far too slow to trace this loop; the sawdust moves because the water bulk-moves. And heated water becomes lighter, so it rises at the heated edge, not sinks.

26. (D) Warm air rises. A heater mounted high near the ceiling warms air that is already at the top, where it tends to stay, leaving the lower (living) part of the room cold. A floor-level heater sends warm air up and around the whole room as a convection loop, so low placement heats a room better. Warm air does not fall down to fill the room; that reverses convection. Air conducts heat poorly, so heating does not spread by conduction regardless of height. And with a low heater the warm air rises, so the floor does not become the hottest part.

27. (C) Convection works only in fluids (liquids and gases) because the warmed parts must rise and circulate, carrying heat bodily. In a solid the particles are locked in their

positions and cannot flow, so no convection current can form — heat spreads through a brick by conduction instead. Paint colour affects radiation, not whether a solid can convect. No amount of heat lets a rigid solid's particles circulate. And solids certainly can be heated in the middle; they just transfer that heat by conduction.

28. (A) Convection is driven by warm (lighter) fluid rising and cold (denser) fluid sinking. Heating at the top leaves the warm water already on top and the cold below — which is already the stable layering — so there is no tendency to overturn and circulate, and heat can only creep downward by slow conduction. Warm water is lighter, not denser, so heating the top does not send it to the bottom. Convection can certainly occur in a beaker when it is heated from below; shape is not the issue. And the cold layer does not 'block' cooling — there is simply no driving force for a current, which is exactly why kettles are heated from below, not above.

29. (B) The dark ploughed field absorbs more sunlight and becomes hotter than pale or shaded surfaces; it warms the air just above it more strongly, and that hotter air expands, becomes lighter, and rises as a vigorous convection current — the thermal the pilot wants. A shaded pond stays cooler and water is not 'always warmest.' A white concrete lot reflects much of the sunlight and warms the air less. And the surfaces do not warm equally despite equal sunlight, because dark surfaces absorb more of it — which is the whole point.

30. (A) The hot air rises away from you, so convection cannot be warming your face. The warmth that crosses the gap — and even passes through a sheet of glass — is radiation: it travels in straight lines, needs no medium, and transparent glass lets much of it through. Convection is ruled out because the hot air goes up, not around the glass. Air is a poor conductor and the gap is large, so conduction is negligible — a chain of touching air does not deliver this warmth. And the heater does not warm you by shoving cold air through the glass; the radiation does.

31. (B) Every warm object emits radiation, and the hotter it is, the more it emits each second. The same poker at 900 °C (bright orange) is hotter than at 600 °C (dull red), so it radiates far more heat per second when bright orange. A cooler object does not shed heat faster — it emits less. It is the same poker, but its temperature is not the same at the two stages, and temperature is the deciding factor. And objects radiate their own heat regardless of sunlight, so the poker radiates whether or not it is in the Sun.

32. (C) With the air pumped out of the chamber there are almost no particles to touch (so practically no conduction) and no fluid to flow and circulate (so no convection); the thin insulating thread carries away almost nothing. Radiation, however, needs no medium and travels through empty space, so the sample sheds heat by radiating it away. There is no gas to carry convection. The vacuum cannot conduct. And it is wrong that no heat can leave — radiation works precisely where the other two fail, which is why hot objects still cool in a vacuum.

33. (A) A dull, dark surface is a good emitter of radiation, so the matt-black mug P radiates its heat away faster and its coffee ends cooler. A shiny foil surface is a poor emitter, so mug Q holds its heat longer — it cools slower, not faster. Equal starting coffee does not make their cooling equal, because the surface type changes how fast each radiates. And objects radiate their own heat regardless of sunlight, so they lose heat by radiation indoors too.

34. (A) A dull dark surface is good at both emitting and absorbing radiation; a shiny surface is poor at both. So the black tank absorbs sunlight faster (heating quicker in the morning Sun) and emits its heat faster (cooling quicker after dark). One property — dark surfaces are strong emitters and absorbers — covers both facts. Shiny surfaces are poor, not good, at both, so they cannot explain it the reverse way. Dark surfaces both absorb and emit well, not one without the other. And colour affects emission too, not absorption alone, so the cooling difference is fully explained by surface type.

35. (C) The gap in a vacuum flask has no particles in contact (so conduction fails) and no fluid to flow (so convection fails). Radiation, needing no medium, crosses freely — so exactly one mode, radiation, can carry heat across the gap. Conduction does NOT cross a vacuum, so 'two' is wrong. Not all three work — only radiation does (which is why the flask's walls are also made shiny, to fight that one remaining path). And 'none' wrongly denies radiation, which genuinely does cross empty space.

36. (A) A good handle must NOT pass heat to the hand, so it should be a poor conductor (insulator) — cork fits. Brass, copper, and steel are all metals and good conductors, so any of them would carry the soldering iron's heat to the grip and burn the hand. Being a good conductor (brass) makes the handle hotter, not coolest. Steel, though a metal, conducts heat well — 'metals barely conduct' is false. And comfort or durability does not override the fact that a copper handle would get dangerously hot.

37. (A) Conduction needs touching particles and convection needs a flowing fluid, so the thin near-still-air gap between the panes greatly slows both of those paths (and a true vacuum gap would stop them almost entirely). Radiation, however, crosses such a gap, so it must be fought separately — by the shiny low-emissivity coating, since shiny surfaces are poor emitters of radiation. Reversing the two (gap stops radiation, shiny stops conduction) is backwards. The gap alone cannot stop radiation, so it does not block all three. And the shiny coating does not block conduction or convection, so it does not handle all three either.

38. (C) Insulation depends on how poorly the wall conducts. The foam box is full of trapped air pockets, the poorest conductor, so it keeps the soup hot longest. The aluminium can conducts the soup's heat outward fastest, so it cools the soup soonest. The glass tumbler sits between them. Metal does not 'seal heat in' — it conducts heat out quickly, the opposite. Same soup and room do not equalise them, since the wall materials

conduct very differently. And trapped air in foam is a BETTER insulator than a solid glass wall, so foam beats glass.

39. (B) Air's widely spaced particles conduct heat very poorly, so when air is trapped and held still (as in a knitted jumper) it barely lets heat through — an excellent insulator, which is why the standing cyclist feels warm. But when air moves (the wind of riding fast), it carries heat away bodily by convection, so it no longer insulates and instead cools the rider. So the jumper's air is not always a perfect insulator — only when still. Moving air does not conduct better; it removes heat by convection, a different mechanism. And air neither destroys nor creates heat; it only transmits or carries it.

40. (D) A thin shiny sheet has almost no trapped air, so it cannot insulate the way wool does. Instead, its shiny surface is a poor emitter and a good reflector of radiation, so it reflects the runner's own radiated body heat back toward them, slowing radiative heat loss. The blanket does not generate warmth — no material makes heat from being touched. A thin sheet does not trap a thick air layer, so that is not its mechanism. And 'conducting the cold away' is backwards: there is no flow of cold, and conducting heat outward would cool the runner.

41. (C) A dark surface is a strong absorber AND a strong emitter of radiation. By day it absorbs more sunlight, so the black shirt feels hotter; at night, facing away from the fire, that same strong-emitting black surface radiates the wearer's body heat away faster. One property explains both. A surface does not switch from absorbing to reflecting by time of day — its absorbing/emitting nature is fixed. White does not absorb more by day; it reflects more. And the night effect is from emission, not cloth thickness, since the shirts are identical except for colour.

42. (D) The job of the bag's surface is to take in as much solar radiation as possible and pass that heat to the water. Dull dark surfaces are the best absorbers of radiation, so black absorbs the most sunlight and heats the water most effectively. Black absorbs rather than reflects, so 'reflects the most' is wrong. Shiny silver reflects more and absorbs less, so it would be worse, not better — and the choice is about absorption, not cost. And the colour of the bag's surface matters greatly; the Sun's heating depends on how much radiation the surface absorbs.

43. (A) A white surface reflects most of the Sun's radiation, so the white drum's walls absorb less heat and the water inside stays cooler. A dark surface absorbs much more, so the dark brown drum's walls take in more heat and its water gets hotter. 'Dark colours keep things cool by absorbing heat' is self-contradictory — absorbing heat warms the drum. Equal size and thickness do not equalise them, because the surface colour changes how much radiation is absorbed. And white reflects, not absorbs, so the white drum is cooler, not hotter.

44. (B) In direct sunlight a dark hat absorbs more of the Sun's radiation, so the black hat warms the head more by day. The catch is that a dark surface is also a better emitter, so at night (no Sun) it would shed the head's own radiated heat faster — a genuine trade-off. White reflects sunlight, so it absorbs less and feels cooler, not warmer, in the Sun. Black does not 'make heat'; it absorbs the Sun's radiation, so it is not warmer at every time. And colour does affect absorption regardless of equal thickness.

45. (D) A dull dark surface is a strong absorber (so the black-lined cooker got hotter in the sun) AND a strong emitter (so it now radiates its heat away faster in the shade). The black-lined cooker therefore both heated up more and cools down faster. Shiny surfaces are poor emitters, so the silver-lined cooker loses heat slowly, not fastest. Equal size and shape do not equalise cooling, because the surface type sets the emission rate. And no closed cooker keeps heat forever — it always radiates heat away to cooler surroundings.

46. (D) By day the land heats faster than the sea. The air over the warm land rises, and cooler air flows in from over the sea to replace it — that inflow from sea to land is the sea breeze, and it carries the flags inland. It is not a land breeze, and the sea does not heat faster than land by day. The Sun does not physically drag air sideways; uneven heating drives the breeze. And the land is the WARMER surface in daytime, not cooler, so the cause is land heating faster, not staying cool.

47. (D) At night the land loses heat and cools faster than the sea, so the sea ends up warmer. Air rises over the warmer sea and cooler air flows out from the cooled land toward the sea — a land breeze (land to sea), which carries the boats out. It is not a sea breeze, and the sea does not cool faster than land. The land is COOLER than the sea at night, not warmer, so 'land stays warmer' is wrong. And the reversal is caused by the land–sea temperature swap, not by an automatic rule.

48. (A) The breeze is driven by the temperature difference between land and sea. By day the land is warmer (sea breeze); by night the land is cooler (land breeze). Near sunrise and sunset the land and sea are passing through nearly equal temperatures, so the difference shrinks to almost nothing and the driving force fades — winds go quiet and shift. There is no fixed one-hour-stop rule; it depends on the temperatures matching. The Sun is low near the horizon at sunrise and sunset, not overhead. And the temperature difference is smallest, not greatest, at those changeover times — that is exactly why the wind is weak then.

49. (A) Both breezes are the same convection mechanism: air rises over the warmer surface and cooler air flows in from the cooler surface. The only thing that changes between day and night is which surface is warmer — the land by day, the sea by night — so the wind simply reverses direction. The same process runs both ways, confirming the second student. The night breeze is not conduction; air conducts heat poorly, and the breeze is bulk air movement. The Moon does not drive the night breeze; uneven heating and cooling do. And neither breeze is radiation — both are convection of air.

50. (B) Sea and land breezes are convection driven by a temperature difference between land and sea: air rises over the warmer surface and cooler air flows in. If both are at exactly the same temperature, no surface is warmer, no air preferentially rises, and so neither breeze can form. The claim that the sea is 'always cooler' is false — by night the sea is the warmer one. A coast does not always have a breeze; without a temperature difference there is none. And the overcast day produces a land breeze only if the land cools below the sea — with no difference, even that does not occur.

51. (B) Warm air rises over the warmer surface, leaving a gap near the ground that is filled by air flowing in from the cooler surface. So the surface breeze always moves FROM the cooler region TOWARD the warmer one: sea-to-land by day (land warmer) and land-to-sea by night (sea warmer). It does not blow from warm to cool at ground level — that air has risen, not flowed sideways. It is not always sea-to-land; the direction reverses at night. And while warm air does rise vertically, the breeze we feel is the horizontal inflow replacing it, not a purely upward wind.

52. (B) The warmth of clothing comes mainly from trapped still air, which conducts heat poorly. Each puffy quilted pocket traps another pocket of still air, so the quilted jacket holds more insulating air pockets than a flat jacket of the same cloth, slowing heat loss more. Weight does not set warmth — trapped air does. Clothes do not generate body heat, so the quilting does not 'produce' more heat. And conducting heat outward faster would cool the child, the opposite of feeling warm.

53. (A) Down insulates by holding many pockets of still air, which conduct heat very poorly. Fluffing it up maximises those air pockets; squashing it flat squeezes the air out, so there is little trapped air left and body heat escapes quickly. The down does not produce heat, so 'stops producing heat' is wrong. A squashed bag is thinner, not thicker, and the loss is from losing trapped air, not from the down conducting better. And the warmth is from trapped air slowing conduction, not from reflecting body heat.

54. (A) A thick towel is full of trapped still air and is a poor conductor, so draping it around the pot greatly slows the rice's heat from escaping — the best insulation here. A shiny metal sheet is a good conductor and would carry heat away from whatever touches it; it does not 'seal heat in' by blocking it, and it only touches the base. Thin metal does not insulate better than a thick towel — it conducts heat away faster. And the two are not equally effective, since the towel insulates far better than the metal sheet.

55. (B) Heat is kept by poor conductors (trapped air) and by covering openings to stop convection/evaporation from the top. The foam mug with a lid combines foam's trapped air with a lid that blocks heat escaping from the open surface, so it stays hottest. The towel-wrapped ceramic mug gains good insulation from the towel's trapped air despite the conducting ceramic, so it is next. A bare ceramic mug insulates the sides moderately but loses heat from the open top. A bare thin metal mug conducts heat out fastest and is open

on top, so it cools soonest. The mugs do not cool equally, because their walls and coverings differ. And the lid plus foam beats bare ceramic, so 'bare ceramic longest' is wrong.

56. (A) A metal ladle left standing in a boiling pot becomes hot by direct particle contact — conduction (correct). Feeling warm where sunlight comes through a closed window is radiation travelling in straight lines from the Sun, crossing the gap with no moving air needed (correct). Water bubbling and churning as it boils is the bulk movement of heated water rising and cooler water sinking — convection (correct). All three pairings match their definitions, so the count is three. Each of the smaller-count options would require at least one pairing to be wrong, but none of them is.

57. (D) A cold-box drastically reduces all three modes — the thick foam starves conduction and convection of easy paths, and the sealed lid blocks air movement — but it cannot remove every path. Heat still leaks slowly through the foam walls, the lid seams, and every opening, so over hours the ice warms and melts. The box does not make 'cold' of its own to 'run out.' The foam slows heat entry, it does not speed it up. And there is no flow of 'cold' into the box — warming is the slow leak of outside heat into the ice.

58. (C) Step 1: the glowing coals warm the cook's face across a gap without touching — that crossing of a gap, in straight lines and needing no moving air, is radiation. Steps 2–3: the heated air over the coals rises and cooler room air flows in to replace it — bulk movement of a fluid, which is convection. The coals cannot conduct heat across the gap to the face (no continuous touching matter), so step 1 is not conduction. Warm air rising and flowing is not radiation, so steps 2–3 are not radiation. And the moving air is convection, not conduction, since air conducts heat poorly and here it is physically flowing — so 'all conduction' is wrong.

59. (D) A laboratory thermometer has no kink, so its thread moves freely. The warm hands heat the bulb's liquid, which expands and pushes the thread up to about 34 °C; once the hands are removed the liquid cools toward room temperature, contracts, and the thread falls back toward 28 °C. This demonstrates the expand-on-heating, contract-on-cooling behaviour the thermometer relies on. Only a clinical thermometer's kink holds the highest reading — this one does not. The thread cannot keep rising once the hands are gone, since no more heat is being added. And with no hand on it the thermometer reads the room's 28 °C, not 0 °C.

60. (B) Both objects are at 5 °C, so a thermometer reads 5 °C on each — they are equally cold by measurement. But marble (stone) is a good conductor and pulls heat from the warm foot rapidly, chilling the skin fast, so it FEELS far colder; the wood, a poor conductor, draws heat slowly and feels less cold. Wood does not feel colder, and it is not at a lower temperature. Feel and temperature do not always match — that mismatch is the whole point, since the marble feels colder despite equal temperature. And the

thermometer would NOT read lower on the marble, because the marble is not genuinely colder — it only conducts heat away faster.